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mcsm-benchs: Benchmarking methods for multi-component signal processing

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Summary

Time-frequency (TF) representations are natural encodings of non-stationary time series, also termed signals, helping to discern patterns that reveal their time-varying frequency structure (Flandrin, 1998). The model one usually has in mind, when discussing TF representations is a so-called multi-component signal (MCS), where the latter is thought to be the sum of several components of individual interest.

MCS processing is an area with a long history which, however, remains a field of methodological innovation (Bardenet et al., 2020; Bardenet & Hardy, 2021; Colominas et al., 2020; Ghosh et al., 2022; Krémé et al., 2020; Legros et al., 2024; Legros & Fourer, 2022; Pascal & Bardenet, 2022a, 2022b). In such a context, systematically comparing existing methods while keeping a record of how they performed on a predefined set of representative problems, i.e. benchmarking, can shed light on novel avenues of research and set clear baselines for new approaches. This is a widely adopted strategy in neighboring fields such as optimization (Bartz-Beielstein et al., 2020; Hansen et al., 2021; Moreau et al., 2022) and machine learning (Mattson et al., 2020), yet there are no benchmarking tools for MCS processing. `mcsm-benchs` is an open-source Python library for creating reproducible benchmarks of methods that aim at extracting information from MCSs. `Benchmark` objects can be created by simply passing a series of simulation parameters, a dictionary of methods and a set of performance metrics, while a `SignalBank` instance can be used to standardize comparisons across benchmarks. The `SignalBank` class can synthesize several MCSs as objects from a custom `Signal` class that behave like regular arrays in Python but also contain information about the components.

Additionally, `mcsm-benchs` includes a `ResultsInterpreter` class that can produce human-readable reports, the functionality of which underpins *collaborative* benchmarks (Moreau et al., 2022). These are based on online repositories (see here for an example) and can be periodically updated by members of the research community, fostering open and collaborative science. Several examples are given in the documentation, as well as a GitHub repository template that relies heavily on `mcsm-benchs` and continuous integration/deployment workflows, in order to automatize the process of publishing new benchmarks.

Statement of need

`mcsm-benchs` brings to the table a common framework to easily carry out extensive comparisons between MCS-based approaches in a unified and objective way. It can be used to benchmark any number of approaches and create clear baselines for new methods that are accessible to the whole research community.

The toolbox is versatile enough to allow comparisons between many kinds of methods. For instance, `mcsm-benchs` was used to compare statistical tests for signal detection (Juan M. Miramont et al., 2022), and for denoising of synthetic and realistic signals under different scenarios, such as white noise or even real-world noises (Juan M. Miramont et al., 2024). As another example, many approaches within MCS processing focus instead on retrieving individual components and estimating their instantaneous frequencies. These methods can be easily benchmarked using `mcsm-benchs` as well (see Juan M. Miramont et al. (2023)).

While the aforementioned cases illustrate the most typical applications in MCS processing, methods are in fact always treated like *black boxes* by `mcsm-benchs`. The only constraint imposed by the software is that the outputs of a method should match the inputs of the performance metrics given by the user. Thanks to this feature, `mcsm-benchs` has a significant potential, as it could be used to systematically compare any signal processing algorithm. Large studies of methods for specific applications can thus be created and kept updated by the signal processing community using `mcsm-benchs`, hopefully leading to widely adopted procedures for evaluating new approaches that are transparent, less time-consuming and straightforward for researchers to use.

Finally, in order to ease the adoption of this package by the community, `mcsm-benchs` also supports methods coded in `Octave`/`Matlab`, so that these can be seamlessly integrated into Python-based benchmarks.

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